VERITAS. Setup Guide

How to download, install and run the project on your own laptop: Windows, macOS or Linux

21CSE306P Applied Generative AI · Team 13 Repository: github.com/OmPathania21/gen_ai_1 (branch **dev**)

Time needed: about 15–20 minutes the first time, mostly waiting for downloads. After that, starting the app takes one command.
Disk space: about 4 GB free. **Internet:** needed for setup, and while using the app (it calls the Groq API).

1. Install the prerequisites
2. Download the project
3. Get your Groq API key
4. Run the setup script
5. Start VERITAS
6. Ask your first question
7. Getting updates later
8. Troubleshooting

> **How to open a terminal. Windows:** press the Windows key, type `cmd`, press Enter (Command Prompt). · **macOS:** press `Cmd + Space`, type `Terminal`, press Enter. · **Linux:** `Ctrl + Alt + T`.
> Type each command exactly as shown and press Enter after each line. Lines in grey starting with `#` are comments: do not type them.

1 Install the prerequisites (once per laptop)

Tool	Version	Where to get it	Check it worked
Git	any	git-scm.com/downloads	`git --version`
Python	**3.10 to 3.13** (3.13 recommended)	python.org/downloads	Windows: `py --version` Mac/Linux: `python3 --version`
Node.js	**20 or newer** (LTS)	nodejs.org (big green "LTS" button)	`node --version`

> **Windows users:** on the first screen of the Python installer, tick **"Add python.exe to PATH"** before clicking Install. If you missed it, run the installer again and choose *Modify*.

After installing, **close and reopen the terminal** so it finds the new tools, then run the three check commands. Each should print a version number.

2 Download the project

Choose a folder to keep it in (for example your Desktop), then run:

WINDOWS

```
cd %USERPROFILE%\Desktop
git clone -b dev
https://github.com/OmPathania21/gen_ai_1.git
cd gen_ai_1
```

MACOS / LINUX

```
cd ~/Desktop
git clone -b dev
https://github.com/OmPathania21/gen_ai_1.git
cd gen_ai_1
```

`-b dev` downloads the **dev** branch, where the code lives. You now have a folder called `gen_ai_1`. **Every command from here on is run inside this folder.**

> **Already cloned it earlier?** Go into the folder and run `git checkout dev` then `git pull`.

3 Get your free Groq API key

VERITAS writes its answers with the **gpt-oss-120b** model, which runs on Groq's free cloud service. Each teammate creates **their own** key; it is free and needs no card.

1. Open **https://console.groq.com/keys** in your browser.
2. Sign in (the *Continue with Google* or *GitHub* button is quickest).
3. Click **Create API Key**. Type any name, e.g. `veritas`, and click **Submit**.
4. **Copy the key immediately.** It looks like `gsk_AbCd1234…` and is shown only once. Paste it into a notes app for the next step.

> **Keep the key private.** Do not send it in WhatsApp groups, paste it into code, or commit it to GitHub. It is stored only in a file called `.env`, which Git is set to ignore. If a key leaks, delete it on the same Groq page and create a new one.

4 Run the setup script (once)

One script does the whole installation. Run it from inside the `gen_ai_1` folder:

WINDOWS

```
py scripts\setup.py
```

MACOS / LINUX

```
python3 scripts/setup.py
```

It prints 7 numbered steps:

Step	What it does	What you do
1/7	Checks your Python version	Nothing
2/7	Creates a private Python environment (`.venv`)	Nothing
3/7	Installs the Python packages (PyTorch, Transformers, ChromaDB…)	Wait: **5–10 minutes**, about 2 GB
4/7	Asks for your Groq API key and saves it to `.env`	**Paste your key and press Enter**
5/7	Builds the web interface (React)	Nothing
6/7	Downloads the 7 demo research papers and the SciFact benchmark	Nothing
7/7	Downloads the local models: MiniLM, ms-marco re-ranker, DeBERTa-v3 NLI (~1 GB)	Wait

> **At step 4 the key is invisible as you paste.** This is normal: the input is hidden like a password. Paste once (`Ctrl+V`, or right-click in Windows Command Prompt; `Cmd+V` on Mac) and press Enter. If you press Enter without a key, the step is skipped and you can add it later (box below).

It ends with **Setup complete.** If any step fails (for example the Wi-Fi drops), just **run the same command again**: finished steps are skipped.

Adding or changing the key by hand (optional)

1. In the `gen_ai_1` folder, open the file `.env` with Notepad (Windows) or TextEdit (Mac).
   No `.env` yet? Create it by copying the example: `copy .env.example .env` (Windows) or `cp .env.example .env` (Mac/Linux). On a Mac, press `Cmd+Shift+.` in Finder to show hidden files.

2. Find the line `GROQ_API_KEY=gsk_your_key_here` and replace the part after `=` with your key: no spaces, no quotes.

```
GROQ_API_KEY=gsk_AbCd1234yourRealKey
```

3. Save the file, and restart VERITAS if it is running.

5 Start VERITAS

WINDOWS

```
start.bat
```

MACOS / LINUX

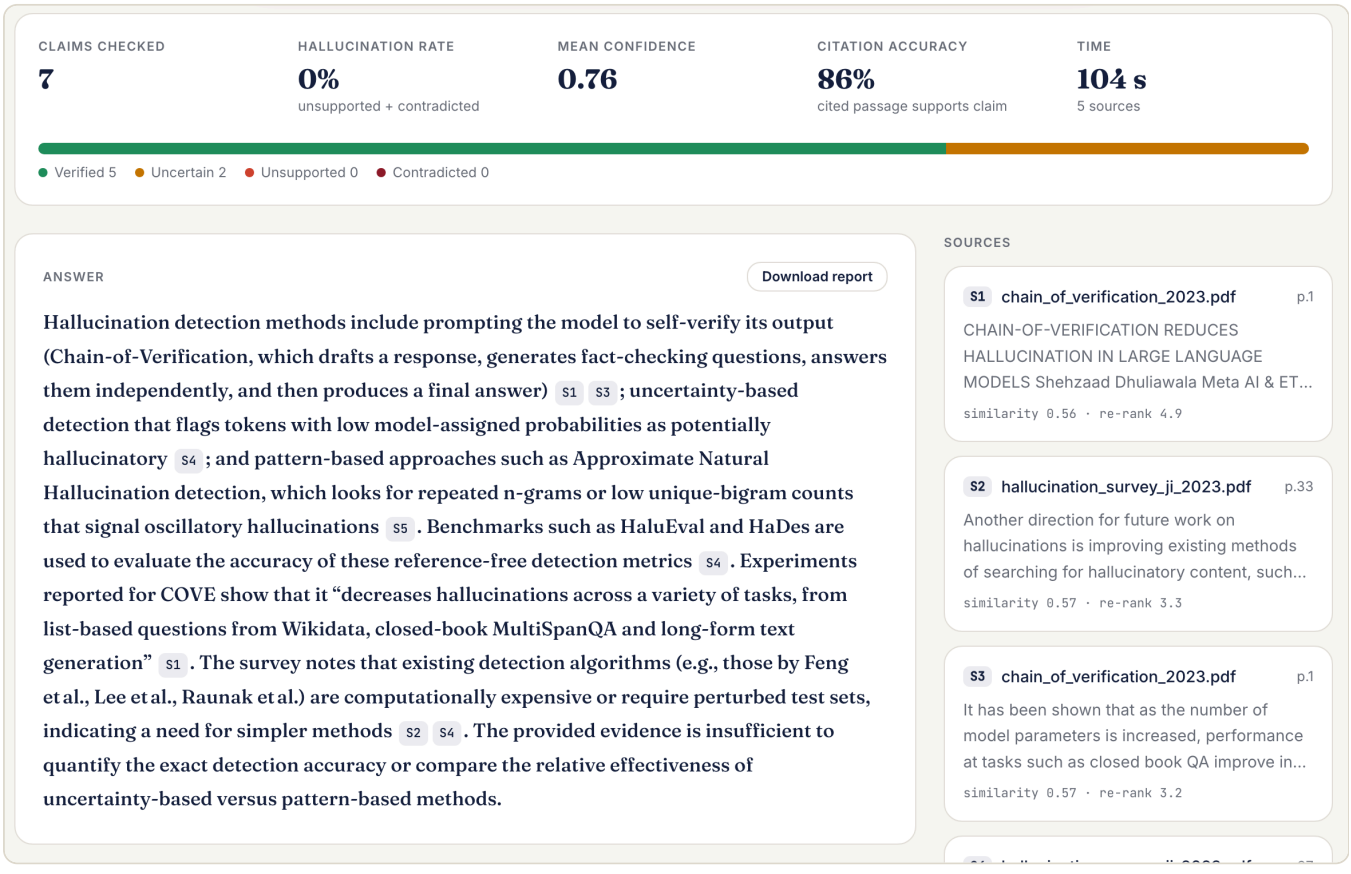
```
./start.sh
```

Wait until the terminal shows `Uvicorn running on http://0.0.0.0:8000`, then open **http://localhost:8000** in your browser.

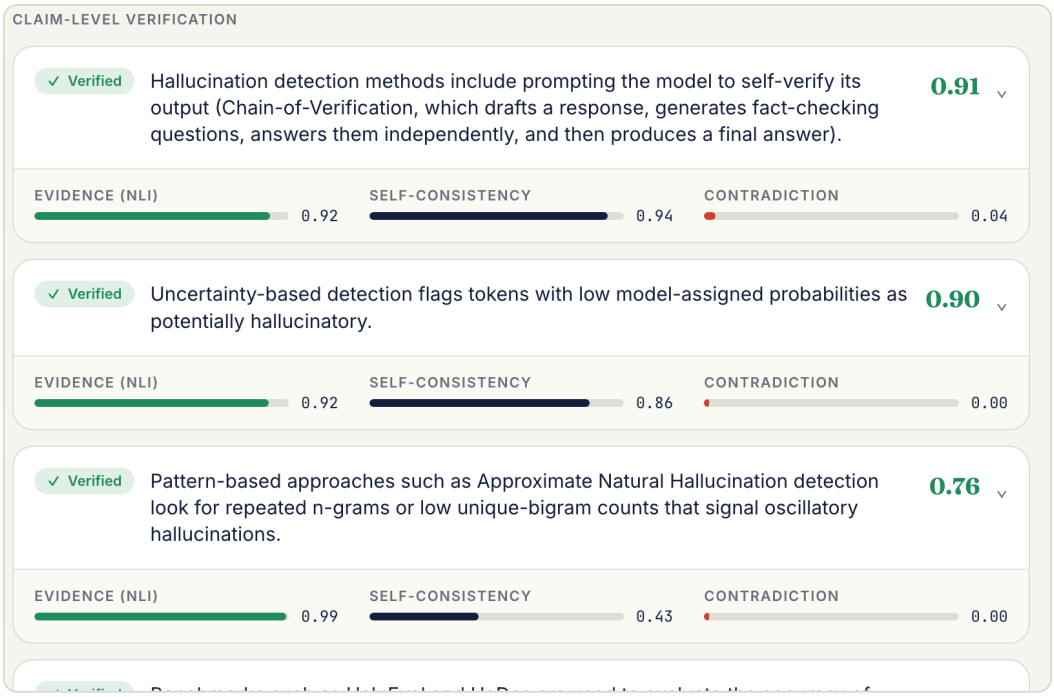
- **Keep the terminal window open** while you use the app: closing it stops VERITAS.
- To stop it, click the terminal and press `Ctrl + C`.
- Next time, you only need this step: open a terminal, `cd` into `gen_ai_1`, and run the start command.

6 Ask your first question

1. On the **Workspace** page, click **Upload a PDF**. Select the demo papers in `gen_ai_1/data/sample_docs/` (you can select all 7 at once), or any PDF, DOCX, TXT or MD file of your own (up to 25 MB).
2. Type a question, or click a suggestion. For the demo papers try: *How can hallucinations in large language models be detected, and how effective are these methods?*
3. Choose **Quick check** (~30 s) or **Deep check** (~1–2 min; also re-asks the model 4 times and corrects weak claims once).
4. Click **Get verified answer**. The first question is slower because the models load into memory.



The answer with [S#] citations, the summary metrics and the source passages



Every claim is checked: Verified / Uncertain / Unsupported / Contradicted, with its confidence score

The other two tabs: **Prompt Lab** compares prompt designs side by side, and **Evaluation** shows the SciFact and prompt-comparison results.

7 Getting updates later

When the code on GitHub changes, stop VERITAS (`Ctrl + C`), then inside `gen_ai_1` run:

WINDOWS

```
git pull
py scripts\setup.py
start.bat
```

MACOS / LINUX

```
git pull
python3 scripts/setup.py
./start.sh
```

Re-running setup takes under a minute, because only new packages are installed. Your key and uploaded documents are kept.

8 Troubleshooting

You see	Fix
`'py' / 'python3' is not recognized` or `command not found`	Python is not installed or not on PATH. Reinstall it (Step 1) and tick *Add python.exe to PATH*. Then close and reopen the terminal.
`'npm' is not recognized` / *Node.js not found*	Install Node.js LTS (Step 1), reopen the terminal, and re-run setup.
`Setup has not been run yet`	Run Step 4 first, from inside the `gen_ai_1` folder.
`No such file or directory` / `pathspec did not match`	You are in the wrong folder. Run `cd` to the `gen_ai_1` folder (it contains `README.md`) and try again.
`GROQ_API_KEY is missing` or the warning *no Groq API key in .env*	Add your key (Step 4, "Adding the key by hand"), then restart VERITAS.
`401` / `Invalid API Key`	The key is mistyped or was deleted. Create a new one (Step 3) and replace it in `.env`.
`429` / `rate limit`	The free tier allows a limited amount per minute. Wait one minute, or use Quick check.
`No documents indexed yet`	Upload at least one document first (Step 6).
`./start.sh: Permission denied` (Mac/Linux)	Run `chmod +x start.sh` once.
`address already in use` (port 8000)	VERITAS is already running in another window, so close it. Mac/Linux alternative: `PORT=8001 ./start.sh`, then open http://localhost:8001.
A package fails to install in step 3/7	Usually a Python version that is too new. Install Python 3.13, delete the `.venv` folder, and re-run setup.
Answers are slow	Normal on laptops without a GPU (Deep check can take 2–3 minutes). An NVIDIA GPU or Apple-silicon Mac is used automatically.

Quick reference

Task	Windows	macOS / Linux
Download (once)	`git clone -b dev https://github.com/OmPathania21/gen_ai_1.git`	
Install (once)	`py scripts\setup.py`	`python3 scripts/setup.py`
Start	`start.bat`	`./start.sh`
Open the app	**http://localhost:8000**	
Stop	`Ctrl + C` in the terminal	
Update	`git pull`, then re-run the install and start commands	
Your key lives in	`gen_ai_1/.env` (never commit or share it)	